ModelIt!

Teacher's Guide

How Do Wildfires Reshape Ecosystems?

Modeling Fire Ecology, Ecological Succession, and Climate-Fire Interactions

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-LS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS2-2

Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to quantify how fire intensity, soil health, and recovery dynamics interact to determine ecosystem outcomes under different fire management and climate scenarios.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Ecosystems are dynamic and subject to natural disturbances like fire that can be essential for maintaining health and diversity. When disturbance regimes are altered beyond historical ranges, ecosystems may shift to fundamentally different and less diverse states.
Crosscutting Concept	**Stability and Change** — Students analyze how fire-adapted ecosystems maintain stability through regular natural disturbance, how disrupting the disturbance regime creates instability, and how climate change is shifting the balance from stabilizing fire to destabilizing megafire.

CAST Testing Alignment

MULTIPLE CHOICE:

A fire-suppressed forest has 8x the natural fuel load. A wildfire ignites during extreme drought conditions. Which outcome does the model predict? A) The fire burns at low intensity because the trees are large and healthy B) The fire burns at extreme intensity, sterilizing the soil and killing seed-producing trees, potentially preventing forest regeneration C) The thick fuel slows the fire spread D) Fire-adapted trees survive regardless of intensity

CONSTRUCTED RESPONSE:

Using your fire ecology model, explain why a forest managed with regular prescribed burns is more likely to survive a wildfire than a fire-suppressed forest, even under identical climate and weather conditions. Reference at least three components including Fuel Load, Fire Intensity, and Soil Health in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Fire has shaped Earth\'s ecosystems for at least 420 million years — since the first land plants created enough fuel and atmospheric oxygen to sustain combustion. Many of the world\'s most biodiverse and productive ecosystems evolved with fire as an essential ecological process: grasslands, savannas, Mediterranean shrublands, boreal forests, and many temperate forests depend on periodic fire to maintain their structure and diversity. But over the past century, a profound shift has occurred: fire suppression policies, climate change, and development in fire-prone areas have transformed fire from an ecological ally into one of the most destructive forces facing natural and human communities.

Fire as an Ecological Necessity

Many plant species are not merely tolerant of fire — they require it. Longleaf pine cannot regenerate without fire to clear competing hardwoods. Giant sequoias need fire\'s heat to open their seed cones. Fireweed and other pioneer species are triggered by fire\'s chemical signals in soil. Fire-adapted grasslands like prairies lose their characteristic plant diversity without periodic burning as woody shrubs invade. The mechanism varies by ecosystem: in frequently-burned systems, low-intensity surface fires clear dead material, recycle nutrients back to the soil, reduce disease and insect populations, and create the open understory conditions that native species need. This is not coincidence — these ecosystems evolved over millions of years with fire as a regular selective pressure, and the species that thrive in them are specifically adapted to the fire regime.

The Fire Suppression Paradox

Beginning in the early 1900s, the US adopted a policy of suppressing all wildfires — the famous Smokey Bear campaign. This policy was remarkably successful at reducing burned acreage: by the mid-20th century, annual US fire area had dropped to a fraction of historical levels. But this success created an invisible crisis. In ecosystems that historically burned every 5-15 years, decades without fire allowed unprecedented fuel accumulation: dense understory growth, dead branches and trees, and ladder fuels that connect ground fire to the canopy. Forests that historically had 20-40 trees per acre now had 200-400. When fire inevitably returned to these fuel-loaded forests, it burned with an intensity that no historical fire ever reached — leaping into the canopy, killing entire forest stands, and sterilizing the soil. The paradox: by suppressing small fires, we guaranteed enormous ones.

Climate Change: Amplifying the Danger

Climate change is transforming fire regimes worldwide through multiple mechanisms. Higher temperatures increase evapotranspiration, drying vegetation and soil faster. Earlier snowmelt extends the fire season by 40-80 days in many regions. Drought intensity and duration are increasing in fire-prone areas. Warmer winters allow bark beetles to survive and kill more trees, adding dead fuel to already overloaded forests. The result is measurable: the area burned annually in the western US has increased by approximately 1,000% since 1970. Global burned area in boreal and temperate forests is at record levels. Climate models project continued increases in fire weather severity throughout the 21st century, with some regions facing fire conditions that have no historical precedent.

Megafires and Type-Conversion

When fire intensity exceeds the evolutionary experience of an ecosystem, recovery may not follow the expected trajectory. Megafires — defined by their extreme behavior, enormous size, and unprecedented intensity — can sterilize soil to depths that destroy seed banks and mycorrhizal networks, create hydrophobic soil layers that prevent water infiltration for years, and kill all seed-producing trees within fire perimeters so large that seeds from surrounding forests cannot reach the center. The result can be type-conversion: permanent ecosystem transformation from forest to shrubland or grassland. This has been documented across the western US, where repeated high-severity fire has converted millions of acres of conifer forest to non-

forest vegetation. Type-conversion represents irreversible biodiversity loss and a shift from carbon sink to carbon source — the forest\'s stored carbon is released by fire and never reabsorbed because the forest does not return.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Climate and Weather Conditions, Fuel Load Accumulation, Fire Intensity, Soil Health and Seed Bank, Post-Fire Vegetation Recovery, Forest Carbon Balance, and Ecosystem Resilience as key components of the fire ecology system.

E **Establish Relationships**

Students determine that fuel load and climate conditions together determine fire intensity, that fire intensity determines whether soil and seed banks survive, that soil and seed bank survival determines recovery trajectory, and that repeated megafires can push ecosystems past resilience tipping points.

V **Visualize & Model**

Students build a computational model showing how fire behavior interacts with ecosystem components to produce different ecological outcomes depending on the intensity and frequency of fire.

E **Evaluate Outcomes**

Students run historical natural fire, suppression legacy fire, and climate-amplified megafire scenarios to test predictions about how fire management and climate change determine ecosystem trajectories.

R **Revise & Extend**

Students add wildland-urban interface density, smoke impacts, or post-fire hydrology to explore how additional factors complicate fire management decisions and affect human communities.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"For millions of years, fire was an ecosystem\'s best friend. Now it\'s one of the most destructive forces on the planet. Today we\'re modeling what changed and what we can do about it."*

Watch For

Show cover image. Quick question: Is fire always bad for forests? Take initial predictions — most students will say yes. By the end of class, they\'ll understand why the answer is much more complex.

Slide 2: Learning Objectives

3 min

Say: *"By the end of today, you\'ll understand a profound paradox: preventing fire is one of the most dangerous things we\'ve ever done to forests."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: fire regime — the pattern of fire an ecosystem evolved with. Disrupting this pattern has consequences.

Slide 3: Big Question

3 min

Say: *"How did something natural become one of the most devastating consequences of climate change? The answer involves a century-long experiment in fire suppression that backfired — literally."*

Watch For

Quick-write: Students write what they think would happen if we stopped fighting all wildfires. Save for comparison — the answer is counterintuitive.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll model fire as an ecological process — not a disaster. This model will reveal why the intensity of fire, not just its occurrence, determines whether it helps or destroys ecosystems."*

Watch For

Preview LEVER steps. Emphasize that fire is a variable, not a binary — low-intensity fire and high-intensity fire produce completely different ecological outcomes.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external drivers of fire behavior, and which are internal ecosystem responses? Think about what sets up fire conditions versus what happens to the ecosystem during and after fire."*

Watch For

Guide sorting: External = Climate/Weather Conditions, Fuel Load Accumulation. Internal = the five ecosystem response variables. Discuss: Fuel Load is external because it\'s determined by management history, but it drives internal fire dynamics.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the cascade from fuel load and climate to fire intensity to soil health to recovery. Where does the system branch between recovery and permanent ecosystem change?"*

Watch For

Students draw arrows. Key discovery: the tipping point where Fire Intensity exceeds Soil Health and Seed Bank survival threshold — this is where the system branches between recovery and type-conversion. Ask: What determines which side of the threshold a fire falls on?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios and watch how dramatically different the outcomes are. The historical natural fire builds ecosystem health. The megafire may destroy it permanently."*

Watch For

Students run scenarios. Key discussion: Compare the natural fire and megafire scenarios — same process (combustion), completely different ecological outcomes. What made the difference? Guide toward fuel load and intensity as the critical variables.

Slide 8: Discoveries

5 min

Say: *"The biggest paradox in fire management: by preventing all fire for a century, we created the conditions for the most destructive fires in history. Let\'s discuss what this means for policy."*

Watch For

Class discussion. Revisit initial predictions. Key challenge: How do you explain to the public that fighting fires can make fires worse? This is a communication challenge as much as a science one.

Slide 9: STEM Challenge

12 min

Say: *"Your community has experienced two devastating megafires in five years. Ecologists say continued suppression will make things worse. Design a plan that protects people while restoring natural fire. Use your model data."*

Watch For

Groups design comprehensive wildfire resilience plans balancing community protection (defensible space, evacuation) with ecological fire management (prescribed burns, fuel reduction). Present with model evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Climate and Weather Conditions, Fuel Load Accumulation
- **INTERNAL:** Fire Intensity, Soil Health and Seed Bank, Post-Fire Vegetation Recovery, Forest Carbon Balance, Ecosystem Resilience

Relationships

- **Fuel Load Accumulation to Fire Intensity:** POSITIVE (+) — More fuel produces more intense fire — dense understory, dead trees, and continuous fuel allow surface fires to ladder into the canopy, transforming manageable surface fires into devastating crown fires.
- **Climate and Weather Conditions to Fire Intensity:** POSITIVE (+) — Hotter temperatures, lower humidity, stronger winds, and drought all increase fire intensity by drying fuel, increasing combustion rate, and driving fire spread. Climate change is amplifying all of these factors.
- **Fire Intensity to Soil Health and Seed Bank:** NEGATIVE (-) — Low-intensity surface fires sterilize only the top centimeter of soil, leaving deeper seed banks and mycorrhizal networks intact. High-intensity fires sterilize to 10+ cm, destroy seed banks, and create hydrophobic layers.
- **Soil Health and Seed Bank to Post-Fire Vegetation Recovery:** POSITIVE (+) — Healthy soil with intact seed banks and mycorrhizal networks supports rapid regeneration — seeds germinate, nutrients are available, and water is retained. Sterilized soil lacks all of these prerequisites for recovery.
- **Post-Fire Vegetation Recovery to Forest Carbon Balance:** POSITIVE (+) — Strong vegetation recovery reestablishes the forest as a carbon sink as growing trees absorb CO2. Failed recovery means the carbon released by fire is never reabsorbed, and the ecosystem becomes a permanent carbon source.
- **Ecosystem Resilience to Post-Fire Vegetation Recovery:** POSITIVE (+) — Resilient ecosystems with intact biodiversity, healthy soil biology, and genetic diversity among tree species recover faster and more completely from fire. Repeated megafires erode resilience, making each subsequent recovery more difficult.

Key Discoveries

- Natural low-intensity fires are essential for ecosystem health — the model shows that ecosystems with regular, moderate fire maintain higher biodiversity, healthier soil, and greater resilience than fire-suppressed ecosystems
- A century of fire suppression has created a dangerous paradox — by preventing small fires that reduce fuel, we have guaranteed large fires that destroy ecosystems, making the forests we tried to protect more vulnerable than they have been in centuries
- Climate change amplifies the fire suppression legacy — hotter temperatures, longer droughts, and lower humidity transform already dangerous fuel loads into conditions for fires of unprecedented intensity that exceed any historical analogue
- High-intensity megafires can push ecosystems past recovery tipping points — when fire sterilizes soil, destroys seed banks, and kills seed-producing trees, the ecosystem may not regenerate as forest at all, instead converting permanently to shrubland or grassland in a type-conversion that represents irreversible habitat loss

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All wildfire is bad and should be prevented"	Fire is an essential ecological process that many ecosystems require for health and regeneration. Fire-adapted species need periodic fire for seed germination, nutrient cycling, and habitat maintenance. The problem is not fire itself but fire of unnatural intensity in fuel-loaded, climate-stressed forests. Low-intensity fire is ecological medicine; high-intensity megafire is ecological catastrophe. The distinction matters enormously because policies that suppress all fire create the conditions for catastrophic fire.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Climate change is the only reason fires are getting worse"	While climate change is a significant amplifier of fire danger, a century of fire suppression is equally responsible for the current megafire crisis. Fire-suppressed forests contain 5-10 times more fuel than historical conditions, creating conditions for unprecedented fire intensity even under historical climate. Climate change and the suppression legacy are compounding factors — each makes the other worse. Addressing only climate (which takes decades) without addressing fuel loads (which can be done now through prescribed burns) leaves communities and ecosystems at extreme risk.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Prescribed burns are too dangerous and should not be done"	While prescribed burns carry some risk, that risk is dramatically lower than the risk of allowing fuel to continue accumulating. Prescribed burns are conducted under carefully controlled conditions — moderate temperature, low wind, adequate humidity, and with trained crews and contingency plans. The risk of a prescribed burn escaping is approximately 1-2%, and escaped prescribed burns are typically small and easily contained. In contrast, the risk of catastrophic wildfire in fuel-loaded forests approaches certainty over time, and megafires kill people, destroy communities, and cause billions in damage. The question is not whether fire will occur, but whether it occurs under controlled conditions or uncontrolled catastrophe.